

pyRTX: a Python package for high precision computation of non gravitational forces on deep space probes

Gael Cascioli^{1,2*}, Ariele Zurria^{3*}, and Erwan Mazarico²

¹ University of Maryland Baltimore County, United States ² NASA Goddard Space Flight Center, United States ³ Sapienza University, Italy * Corresponding author * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- Review
- Repository
- Archive

Editor: [Open Journals](#)

Reviewers:

- @openjournals

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

In partnership with



AMERICAN
ASTRONOMICAL
SOCIETY

This article and software are linked with research article DOI [10.3847/xxxxx](https://doi.org/10.3847/xxxxx) <- update this with the DOI from AAS once you know it., published in the The Planetary Science Journal.

Summary

With the constant improvement of radiometric tracking systems, inaccuracies in the non-gravitational force modeling have become one of the limiting factors to deep space precise orbit determination, and the scientific products that it enables. The main factor impacting the limited accuracy of non-gravitational force models is the complex 3D shape of the spacecraft. While fast, reliable, analytical models are available for simple shapes (spheres, flat plates, etc), no such model is generally available for a complex shape. This software package aims to address this limitation by leveraging ray-tracing to compute the complex interaction between the forcing environment (radiation, atmosphere) and the three dimensional shape of the spacecraft. This software is specifically geared towards planetary OD where inaccuracies of the model often cannot be overcome with continuous tracking as it is routinely done for Earth-orbiting spacecraft.

Statement of need

Several scientific investigations require high-precision reconstruction of spacecraft trajectories. Among these, one of the most demanding is the determination of the gravity field of Solar System bodies (planets, moons). This task is accomplished by solving the so-called orbit determination (OD) problem (Milani & Gronchi, 2009; Tapley et al., 2004). The solution of the OD in the adjustment of a dynamical model (a set of differential equations) describing the spacecraft motion. Systematic errors in the dynamical model will almost inevitably lead to systematic errors in the solution. In the recent years significant improvements in radiometric tracking system, have led to more and more precise measurements of the spacecraft position and velocity (the input to the OD), thus requiring increasingly more accurate dynamical models (Asmar et al., 2005; Cappuccio et al., 2020; Cappuccio et al., 2025; Mazarico et al., 2023). One of the major limitations of current dynamical modelling of deep-space probes consists in the complex interaction between the spacecraft shape and the atmosphere, and with radiative forces (solar radiation pressure, albedo, thermal infrared radiation). We developed the pyRTX software package to address this limitation. Leveraging the ray-tracing technique, originally developed in computer graphics, instead of relying on simplified macro-models (flat plates, cylinders, etc) pyRTX is able to use the actual 3D shape of the spacecraft (provided as, for example, an .obj file), to compute several non-gravitational accelerations.

pyRTX addresses the gap in open source solutions for a comprehensive modelling of several, important, non-gravitational forces. Being built around the NAIF SPICE astrodynamical library (Acton, 1996) and its Python wrapper (Annex et al., 2020), pyRTX is intended to be a plug-in tool that can be used in existing OD codes, and applications.

41 **Functionality**

42 In this section we describe the main functionalities of the pyRTX software. All of these
43 functionalities are discussed in the set of example scripts and notebooks included in the code
44 distribution. Our companion paper (Zurria et al., 2026) discusses an actual application of the
45 pyRTX library for ameliorating the OD of NASA's Lunar Reconnaissance Orbiter.

- 46 ■ *Solar Radiation Pressure Modeling*: pyRTX computes the acceleration due to solar
47 photons by casting rays from a pixel plane representing the incoming solar flux. The
48 engine inherently accounts for self-shadowing (where spacecraft components block light
49 from reaching others) and multiple levels of reflections. Users can specify optical
50 properties for each mesh face, allowing the software to simulate both specular and diffuse
51 (Lambertian) reflections.
- 52 ■ *Planetary Radiation Pressure*: The software models the effects of radiation reflected
53 (albedo) and emitted (thermal infrared) by planetary bodies. This includes the ability to
54 use complex planetary shape models (e.g., topography from digital elevation models,
55 DEMs) and spatially variable maps for albedo, emissivity, and surface temperature.
- 56 ■ *Eclipse and Shadow Function Analysis*: pyRTX can compute precise shadow functions
57 during eclipse transitions. It supports advanced modeling features such as solar limb
58 darkening, where the variation in intensity across the solar disk is accounted for in the
59 flux calculation.
- 60 ■ *Atmospheric Drag*: For low-altitude missions, the software calculates the effective
61 aerodynamic cross-section of the spacecraft, providing inputs for atmospheric drag
62 modeling.
- 63 ■ *Lookup Table (LUT) Generation*: To enable efficient integration with OD software,
64 pyRTX can pre-compute accelerations over a grid of incident directions. This feature
65 supports spacecraft with articulating components (e.g., solar arrays), allowing users
66 to generate comprehensive LUTs that map acceleration vectors to specific spacecraft
67 orientations and strongly reduce computation time.

68 **Acknowledgements**

69 Work by G.C. was supported by NASA under award No. 80GSFC24M0006.

70 **References**

- 71 Acton, C. H. (1996). Ancillary data services of NASA's navigation and ancillary information
72 facility. *Planetary and Space Science*, 44(1), 65–70. [https://doi.org/10.1016/0032-](https://doi.org/10.1016/0032-0633(95)00107-7)
73 [0633\(95\)00107-7](https://doi.org/10.1016/0032-0633(95)00107-7)
- 74 Annex, A., Pearson, B., Seignovert, B., Carcich, B., Eichhorn, H., Mapel, J., Von Forstner, J.,
75 McAuliffe, J., Del Rio, J., Berry, K., Aye, K.-M., Stefkó, M., De Val-Borro, M., Kulumani,
76 S., & Murakami, S. (2020). SpiceyPy: A pythonic wrapper for the SPICE toolkit. *Journal*
77 *of Open Source Software*, 5(46), 2050. <https://doi.org/10.21105/joss.02050>
- 78 Asmar, S. W., Armstrong, J. W., Iess, L., & Tortora, P. (2005). Spacecraft Doppler tracking:
79 Noise budget and accuracy achievable in precision radio science observations. *Radio Science*,
80 40(2), n/a–n/a. <https://doi.org/10.1029/2004RS003101>
- 81 Cappuccio, P., Notaro, V., Ruscio, A. di, Iess, L., Genova, A., Durante, D., Stefano, I. di, Asmar,
82 S. W., Ciarcia, S., & Simone, L. (2020). Report on First Inflight Data of BepiColombo's
83 Mercury Orbiter Radio Science Experiment. *IEEE Transactions on Aerospace and Electronic*
84 *Systems*, 56(6). <https://doi.org/10.1109/TAES.2020.3008577>

- 85 Cappuccio, P., Sesta, A., Di Benedetto, M., Durante, D., De Filippis, U., Di Stefano, I., Iess,
86 L., Mackenzie, R., & Godard, B. (2025). Analysis of radio science data from the KaT
87 instrument of the 3GM experiment during JUICE's early cruise phase. *Aerospace*, 12(1),
88 56. <https://doi.org/10.3390/aerospace12010056>
- 89 Mazarico, E., Buccino, D., Castillo-Rogez, J., Dombard, A. J., Genova, A., Hussmann, H.,
90 Kiefer, W. S., Lunine, J. I., McKinnon, W. B., Nimmo, F., Park, R. S., Roberts, J.
91 H., Srinivasan, D. K., Steinbrügge, G., Tortora, P., & Withers, P. (2023). The Europa
92 Clipper Gravity and Radio Science Investigation. *Space Science Reviews*, 219(4), 30.
93 <https://doi.org/10.1007/s11214-023-00972-0>
- 94 Milani, A., & Gronchi, G. (2009). *Theory of Orbit Determination*. Cambridge University Press.
95 <https://doi.org/10.1017/CBO9781139175371>
- 96 Tapley, B. D., Schutz, B. E., & Born, G. H. (2004). *Statistical Orbit Determination*. Elsevier.
97 <https://doi.org/10.1016/B978-0-12-683630-1.X5019-X>
- 98 Zurria, A., Cascioli, G., & Mazarico, E. (2026). Refining non-gravitational force modelling
99 of the Lunar Reconnaissance Orbiter with ray tracing. *The Planetary Science Journal*.
100 <https://doi.org/10.0000/pending>

DRAFT